Dep… of Electrical Engineering

End Term Exam: April 2026

Course: B.Tech	Semester: IV
Subject: Power Systems	Subject Code: EE24221
Time Duration: 120 Minutes	Max. Marks: 40

Note: All questions are compulsory else do as directed.

S. No	Question	Marks	COs
1	a) Explain the reasons why bundle conductors are preferred in overhead transmission lines? How bundle conductors affect inductance and capacitance of the lines?	4	3
	b) What is role of transposition in overhead transmission lines? What happens if we do not transpose the lines?	3	2
	c) Explain the phenomenon of corona?	3	1
2	Derive the ABCD parameters of π and T models of transmission line.	10	2
3	a) A single phase distributor one km long has resistance and reactance per conductor of 0·1 Ω and 0·15 Ω respectively. At the far end, the voltage VB = 200 V and the current is 100 A at a p.f. of 0·8 lagging. At the mid-point M of the distributor, a current of 100 A is tapped at a p.f. of 0·6 lagging with reference to the voltage V_M at the mid-point. Calculate: (i) voltage at mid-point (ii) sending end voltage V_A (iii) phase angle between V_A and V_B?	5	4
	b) Non-reactive loads of 10 kW, 8 kW and 5 kW are connected between the neutral and the red, yellow and blue phases respectively of a 3-phase, 4-wire system. The line voltage is 400V. Calculate (i) the current in each line and (ii) the current in the neutral wire.	5	4
4	a) Explain the methods to determine the per phase capacitance (C_p) and core-core capacitance (C_c), and core-sheath capacitance (C_s) of three phase underground cables?	5	2
	b) A single-core lead sheathed cable is graded by using three dielectrics of relative permittivity 5, 4 and 3 respectively. The conductor diameter is 2 cm and overall diameter is 8 cm. If the three dielectrics are worked at the same maximum stress of 40 kV/cm, find the safe working voltage of the cable. What will be the value of safe working voltage for an ungraded cable, assuming the same conductor and overall diameter and the maximum dielectric stress?	5	4